

Data Preprocessing Report

Project: Advanced Regression Model Comparison & Optimization Platform
Dataset: House Prices – Advanced Regression Techniques
Target Variable: SalePrice

Dataset Overview

Original Dataset Shape: (1460, 81)
Processed Dataset Shape: (1458, 266)

Missing Value Summary

Feature	Missing Percentage
LotFrontage	259
Alley	1369
MasVnrType	872
MasVnrArea	8
BsmtQual	37
BsmtCond	37
BsmtExposure	38
BsmtFinType1	37
BsmtFinType2	38
Electrical	1
FireplaceQu	690
GarageType	81
GarageYrBlt	81
GarageFinish	81
GarageQual	81
GarageCond	81
PoolQC	1453
Fence	1179
MiscFeature	1406

Duplicate Check

Duplicate Records Found: 0
Action Taken: No duplicate records were removed.

Outlier Treatment

Outliers were identified using the IQR method. Extreme observations in GrLivArea with low SalePrice were removed to improve model performance.

Outlier Summary

Feature	Outlier Count
LotFrontage	93
LotArea	69
MasVnrArea	98
BsmtFinSF1	7
BsmtFinSF2	167
BsmtUnfSF	29
TotalBsmtSF	61
1stFlrSF	20
2ndFlrSF	2
GrLivArea	31
GarageArea	21
WoodDeckSF	32
OpenPorchSF	77
EnclosedPorch	208
3SsnPorch	24
ScreenPorch	116
PoolArea	7
MiscVal	52
SalePrice	61

Feature Engineering

Six new features were created to improve model performance:

- TotalSF
- TotalBathrooms
- HouseAge
- RemodelAge
- LivingSpaceEfficiency
- TotalPorchSF

Encoding Categorical Variables

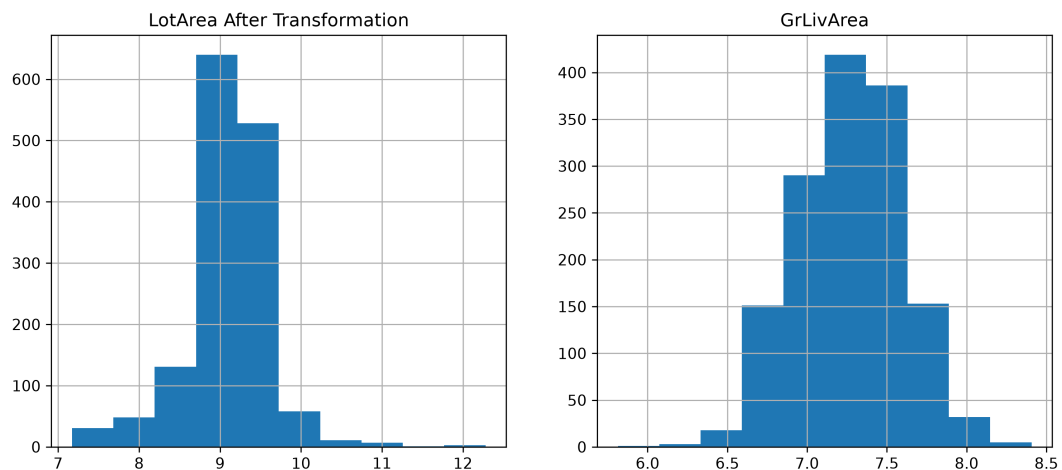
Categorical features were converted into numerical values using One-Hot Encoding. The `drop_first` option was applied to reduce multicollinearity and prepare the dataset for machine learning models.

Scaling Numerical Features

Numerical features were standardized using `StandardScaler` to ensure a consistent feature scale. The target variable (`SalePrice`) was not scaled.

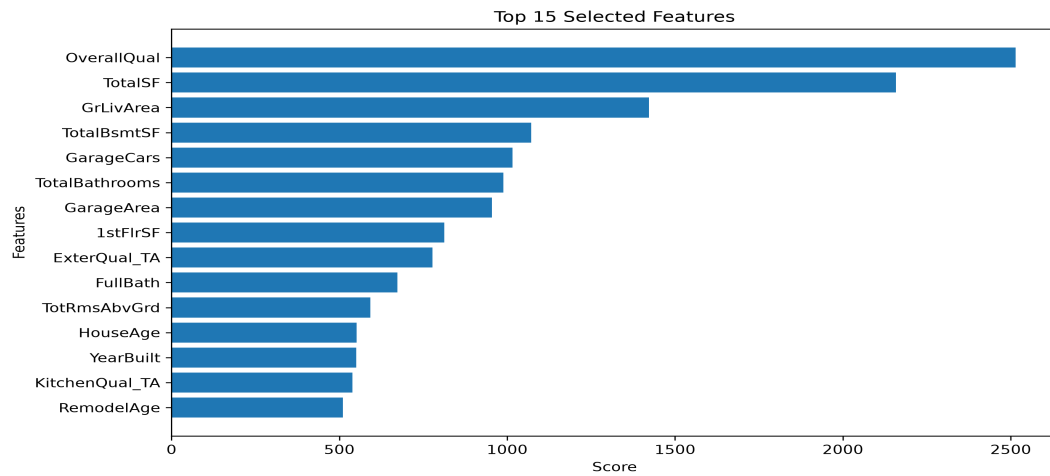
Handling Skewed Distributions

Highly skewed numerical features ($\text{skewness} > 0.75$) were transformed using the `log1p()` function to reduce skewness and improve model performance.



Feature Selection

`SelectKBest` with the `f_regression` method was used to identify the most relevant features. The top 25 features were selected for model training.



Conclusion

The dataset was successfully preprocessed by handling missing values, removing outliers, engineering new features, encoding categorical variables, scaling numerical features, reducing skewness, and selecting the top 25 features. The processed dataset is ready for model development and evaluation.